

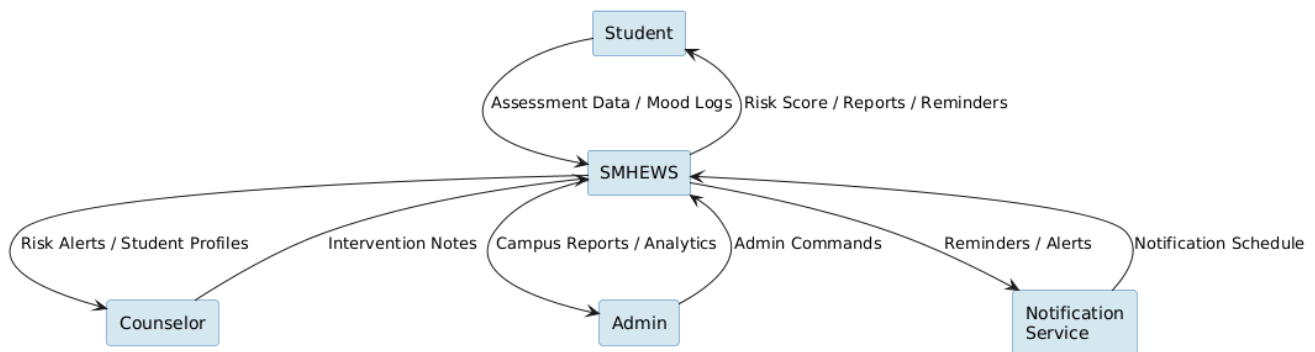
MILESTONE 2: Modelling

3. Data Flow Diagrams

a. Level-0 DFD (Context Diagram)

Shows the entire SMHEWS as a single process with external entities and data flows.

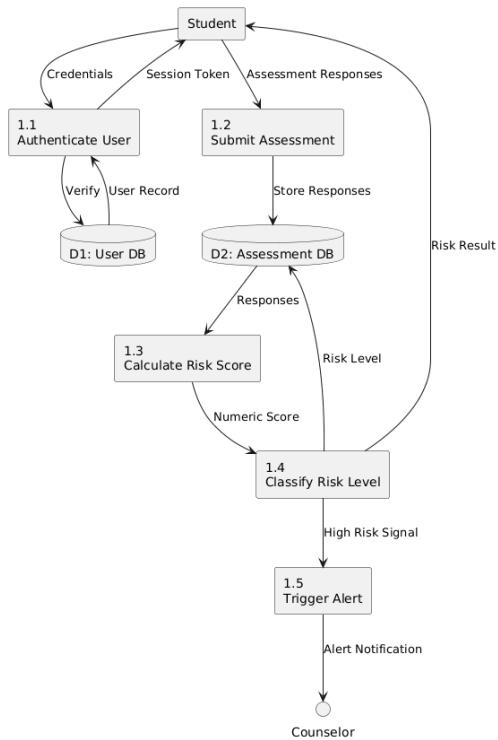
Level-0 DFD



b. Level-1 DFD (Assessment & Risk Processing)

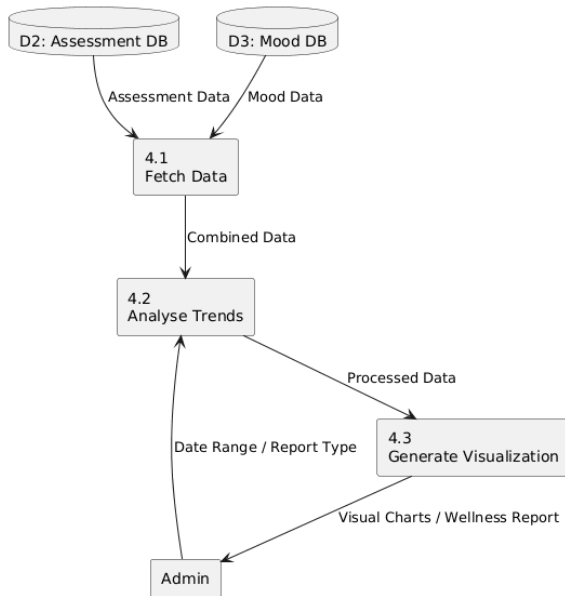
Decomposes the core assessment and risk computation process into sub-processes.

Level-1 DFD (Assessment)



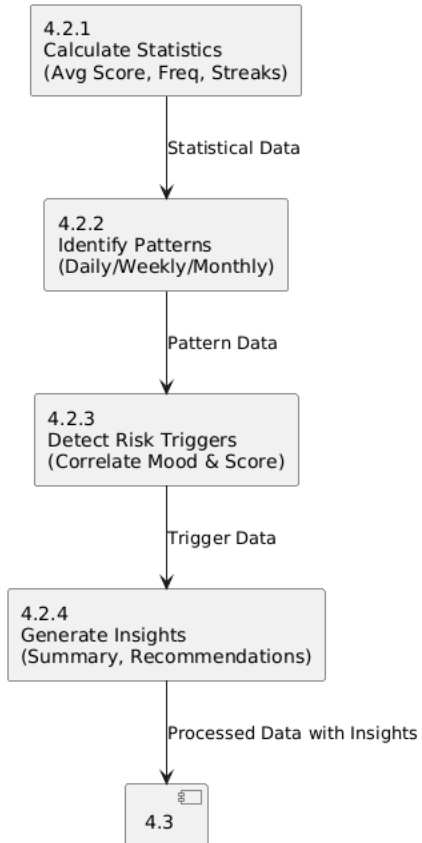
c. Level-1 DFD (Report Generation)

Level-1 DFD (Reports)



d. Level-2 DFD (Analyse Trends)

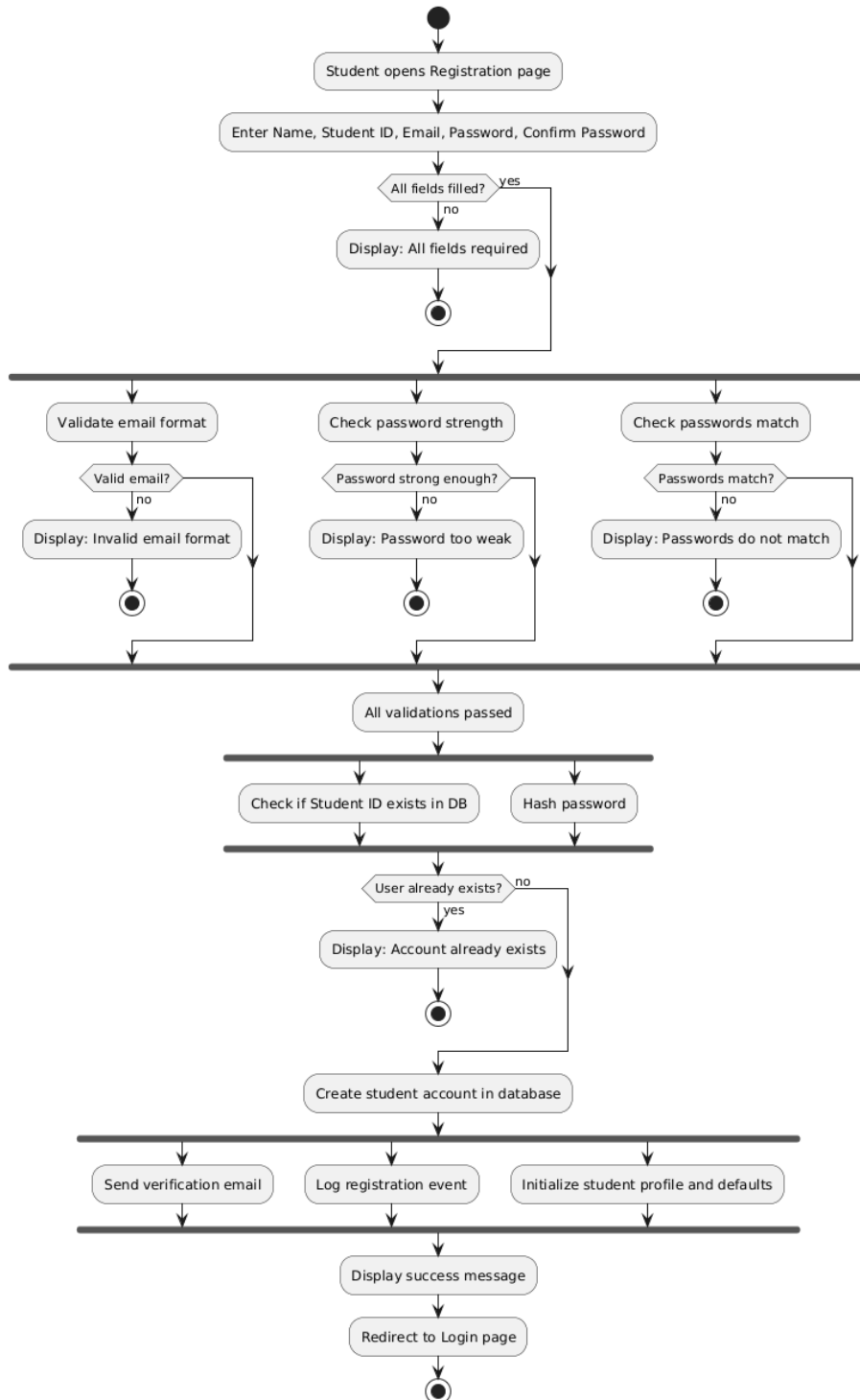
Level-2 DFD



4. Activity Diagrams

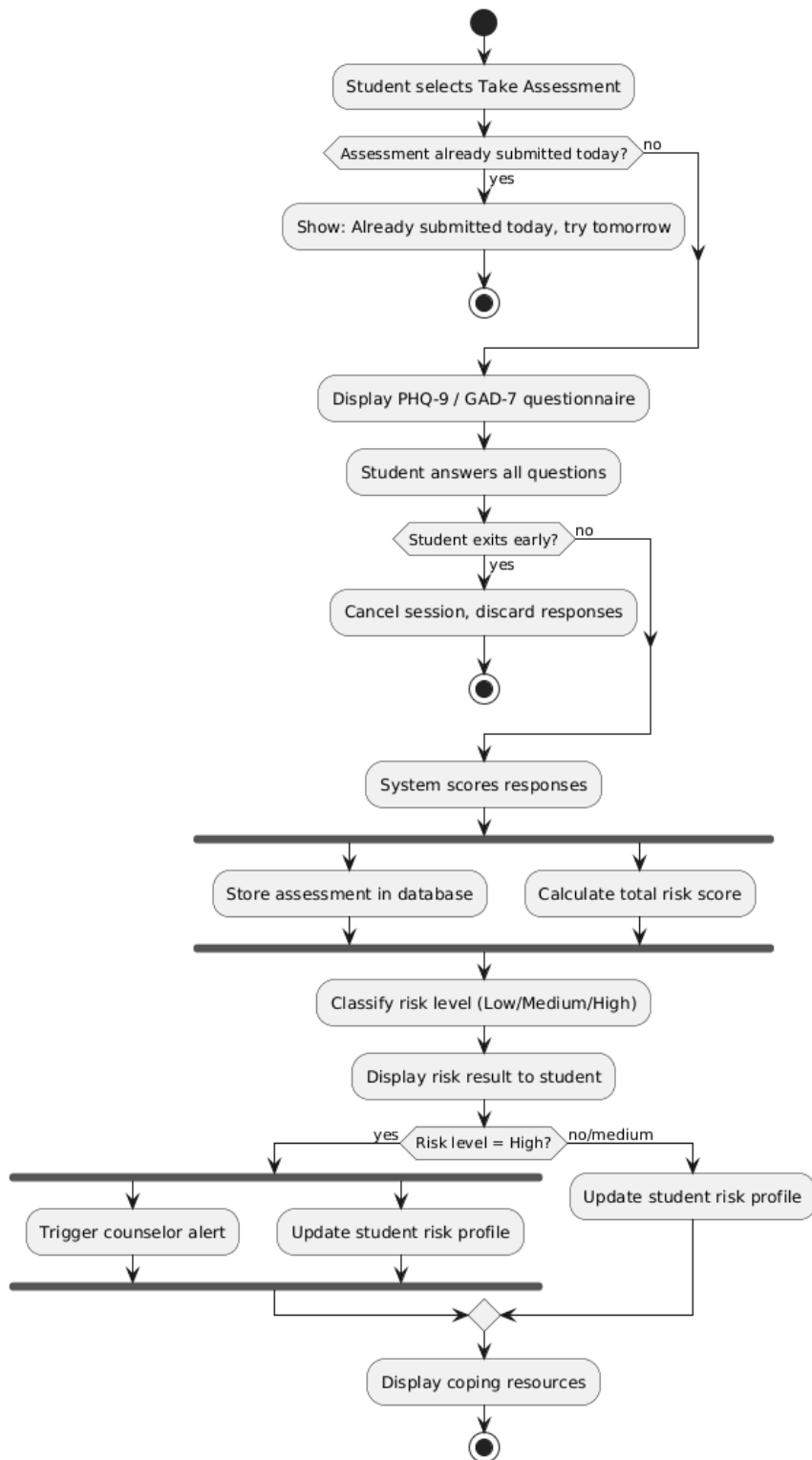
a. Student Registration

Activity: *Student Registration*



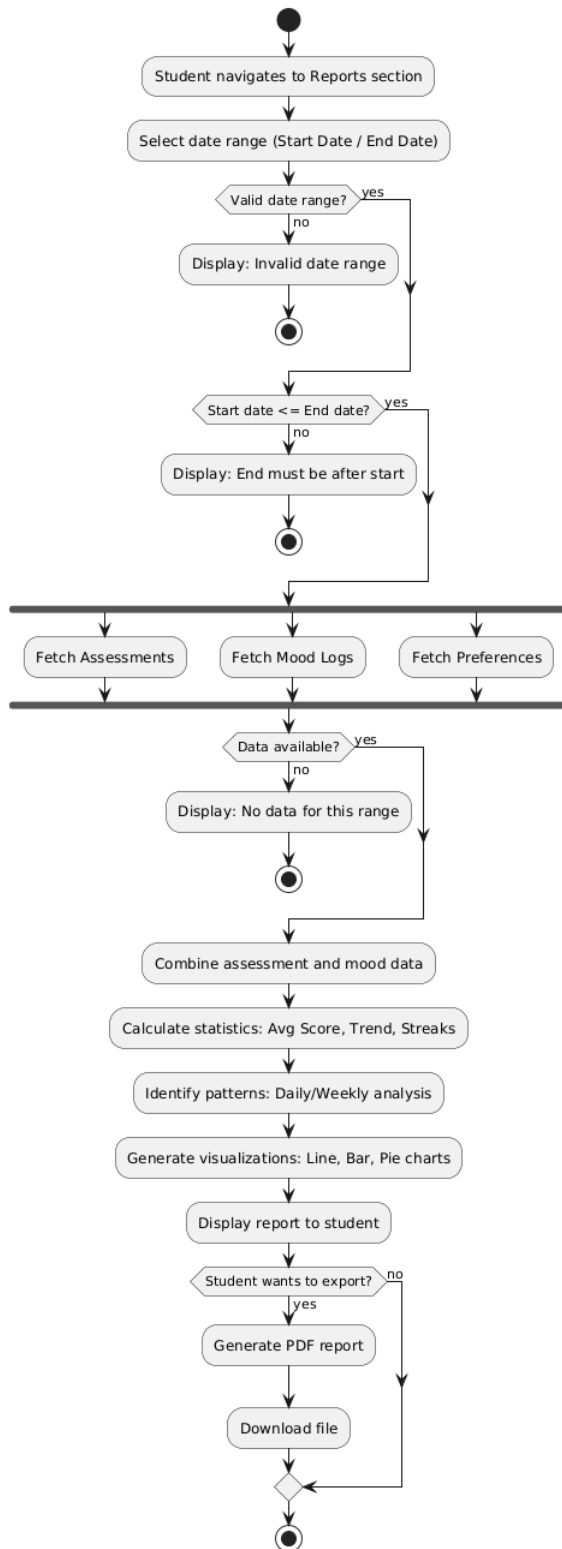
b. Complete Self-Assessment

Activity: Self-Assessment



c. View Progress Report

Activity: View Progress Report



5. Context Diagram

Shows all external entities interacting with the SMHEWS and the nature of data exchanged.

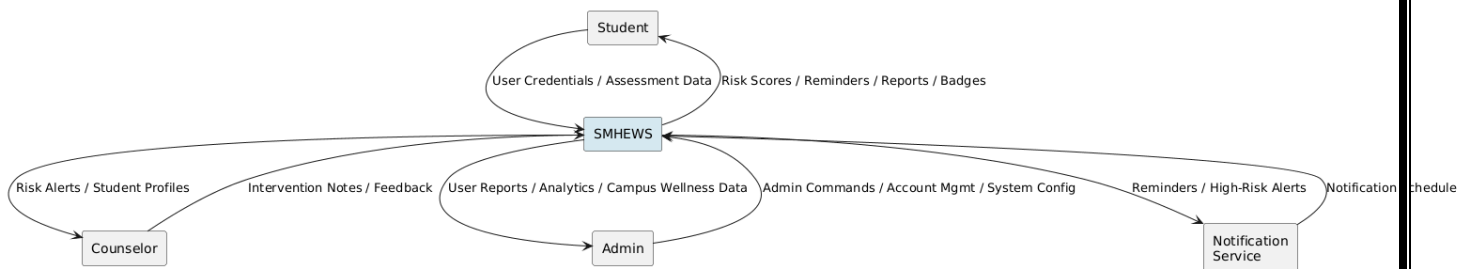
External Entities

- Student – interacts with the system to register/login, complete assessments, log moods, submit anonymous reports, and view wellness reports.
- Counselor – receives risk alerts, views student profiles, and records intervention notes.
- Admin – manages system settings, views campus-wide analytics, and generates aggregate reports.
- Notification Service – delivers daily wellness reminders, assessment prompts, and high-risk counselor alerts.

Data Flow

- Input to System: student credentials, assessment responses, mood logs, anonymous reports, admin commands, notification schedule.
- Output from System: risk scores, wellness dashboards, counselor alerts, campus analytics, PDF reports, push notifications.

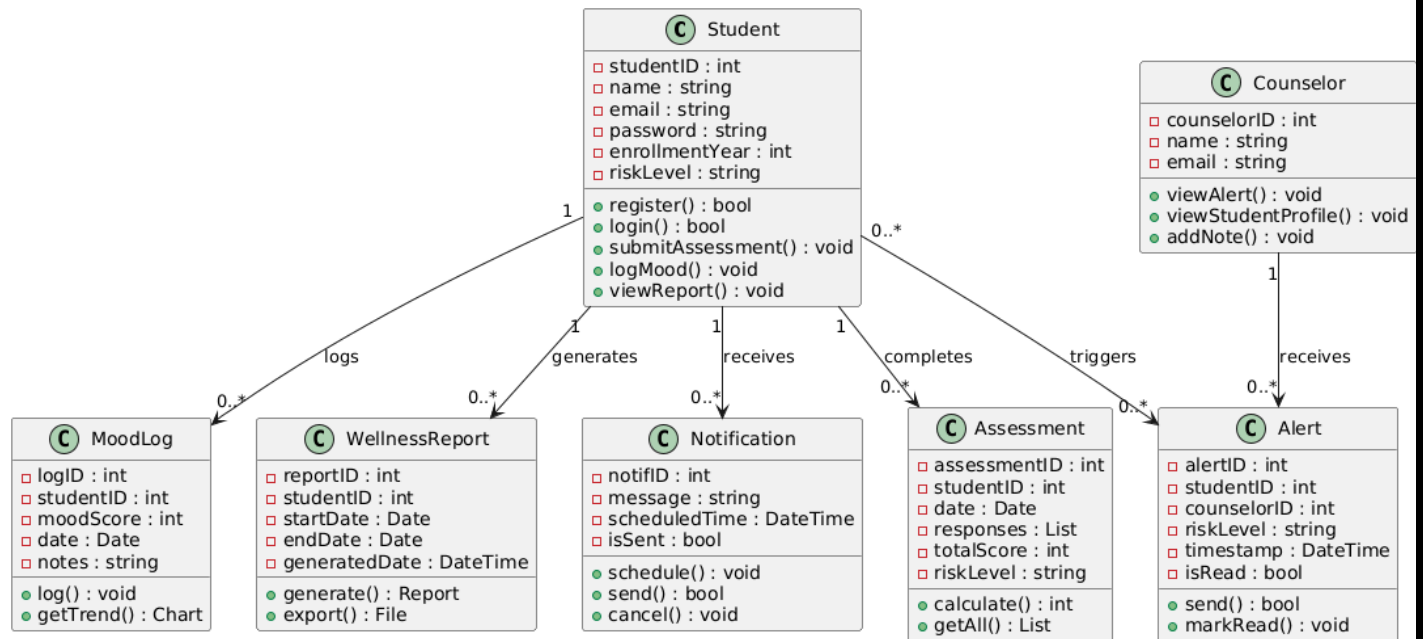
Context Diagram



6. UML Class Diagram

Defines the main classes, their attributes, methods, and relationships within the SMHEWS.

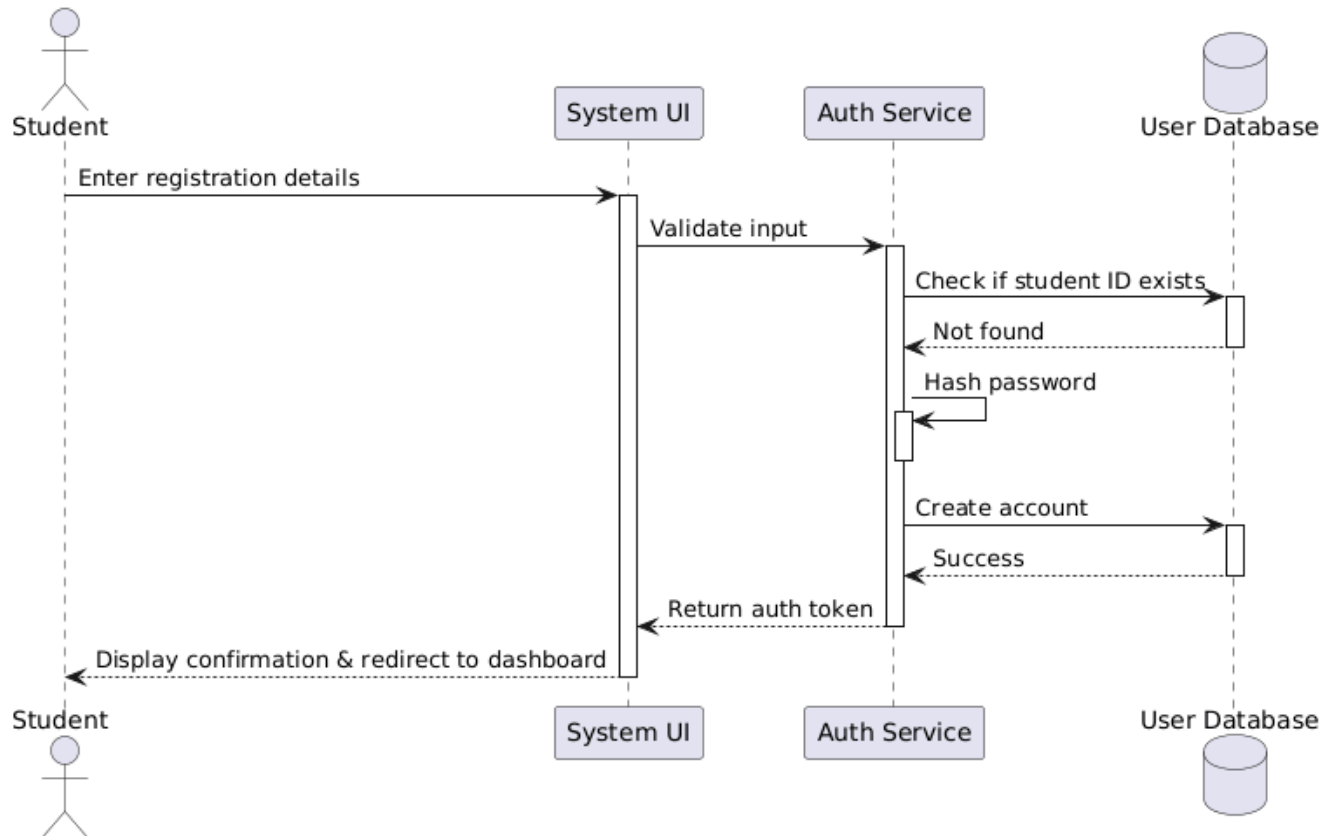
UML Class Diagram



7. Sequence Diagrams

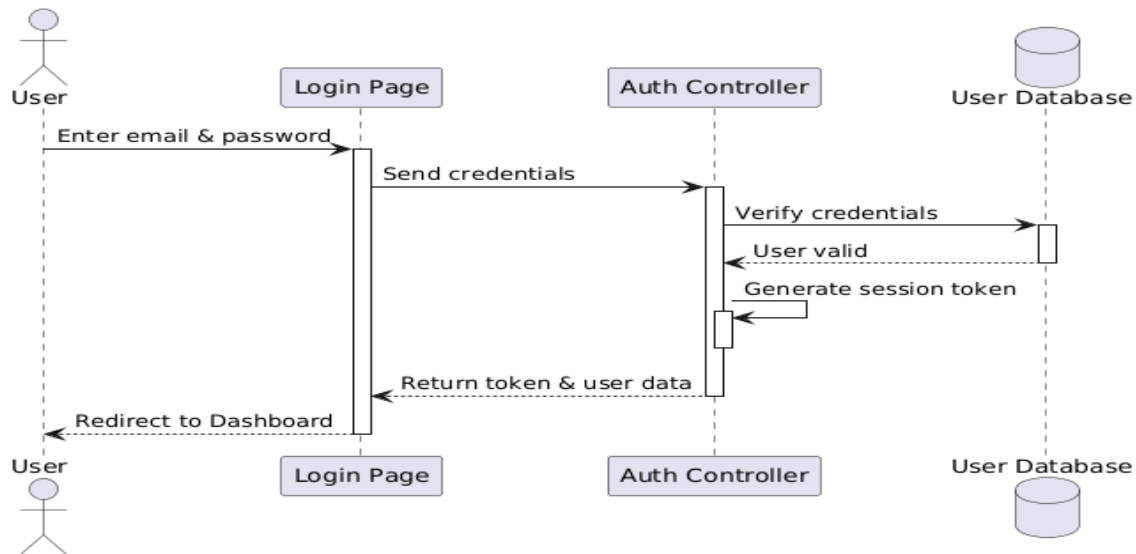
1. User Registration Sequence

Sequence: Registration



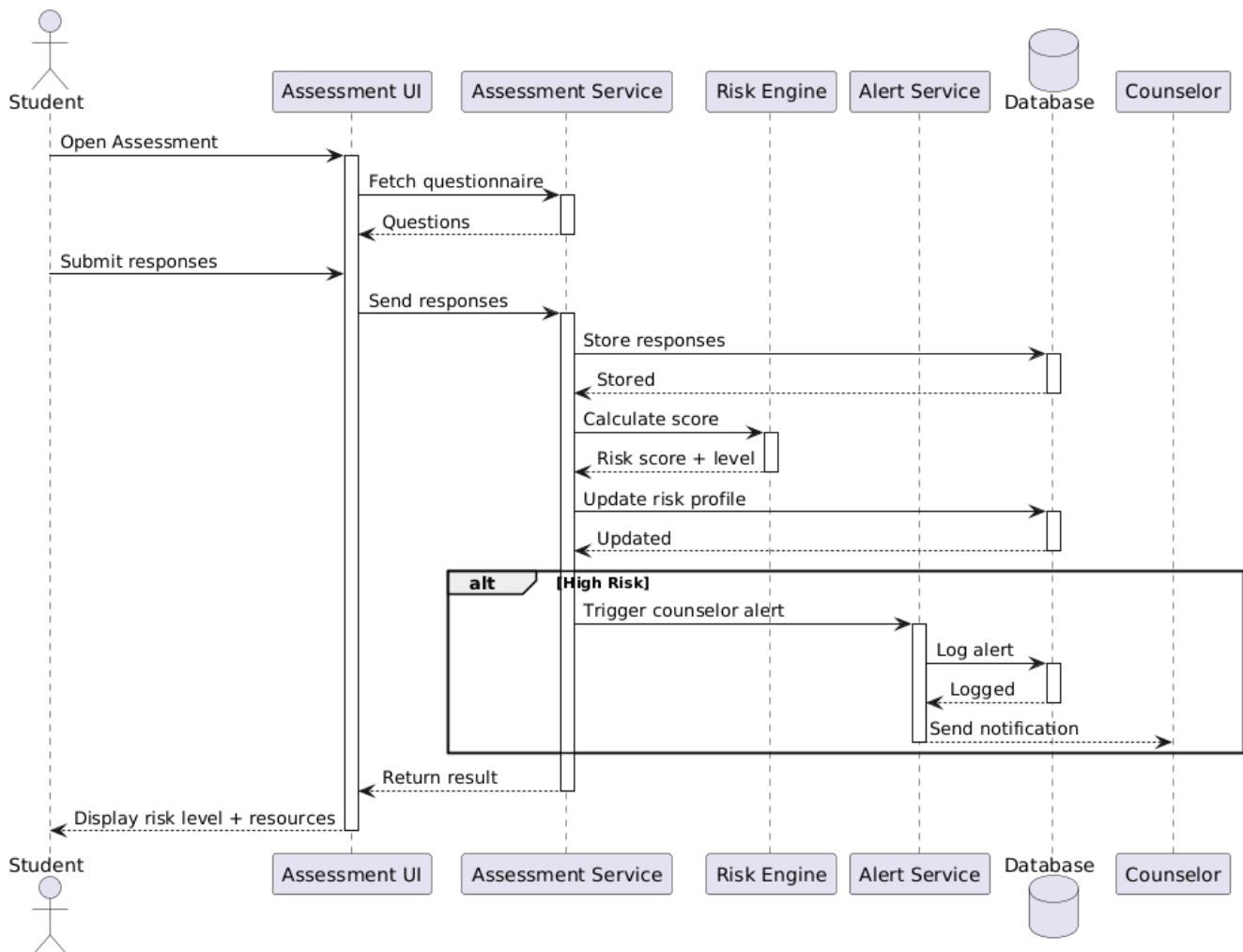
2. Login Sequence

Sequence: Login



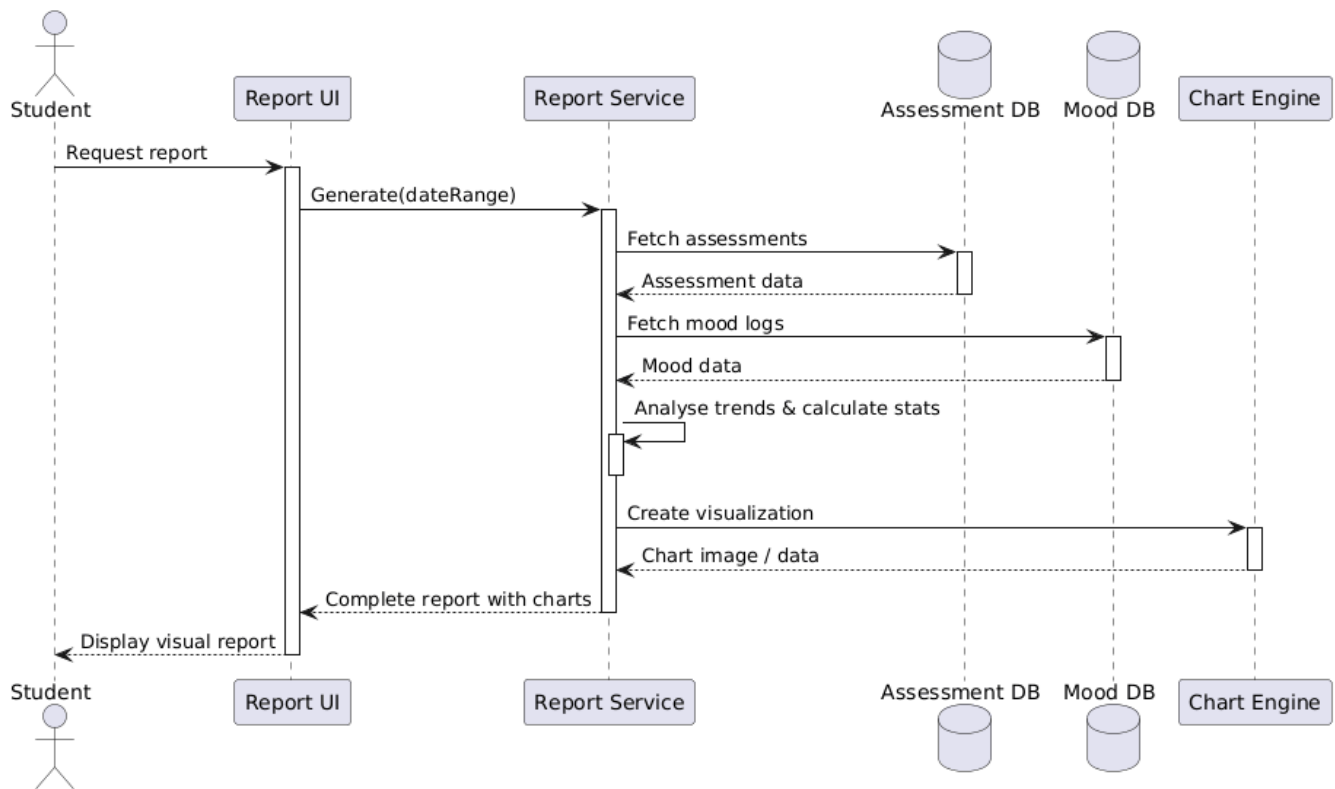
3. Complete Self-Assessment Sequence

Sequence: Self-Assessment



4. Generate Progress Report Sequence

Sequence: Progress Report



5. Send Counselor Alert Sequence

Sequence: Counselor Alert

